1.14

$P(AB)$ is the probability of the interesection of the event sets A and B.

Thus the intersection of the two sets is at most the size of the smaller set. Thus $P(AB)$ is at most $P(A)$

To find the minimum probability, its trying to minimize the intersection of sets A and B. In otherwords, maximumizing the intersection of A and B^c. $P(A) = 0.4$ and $P(B^c) = 1 - P(B) = 1 - 0.7 = 0.3$ The maximum of this intersection is 0.3 because $P(B^c)$ is the smaller one, thus the remaining 0.1 probability must be in the intersection of A and $(B^c)^c = B$. Thus the minimum of $P(AB)$ is 0.1

Therefore $0.1 \leq P(AB) \leq 0.4$

1.18

$\Omega = \{3, 4, 5\}$

k	3	4	5
$p_X(k) = P(X == k)$	$\frac{3}{16}$	$\frac{1}{2}$	$\frac{5}{16}$

1.26

$|\Omega| = \binom{15}{4} = 1365$

$|\text{2 men 2 women}| = \binom{10}{2} * \binom{5}{2} = 45 * 10 = 450$

$P(\text{2 men 2 women}) = \frac{|\text{2m2w}|}{|\Omega|} = \frac{450}{1365} = \frac{30}{91}$

1.28

a

A is w/o replacement

$P(A) = \frac{n^2 - n + m^2 - m}{(n+m)(n+m-1)}$

Alternatively, $P(A) = 1 - \frac{n*m}{(n+m)(n+m-1)}$

b

B is w/ replacement

$P(B) = \frac{n^2+m^2}{(n+m)^2}$

Alternatively, $P(B) = 1 - \frac{n*m}{(n+m)^2}$

c

Unless $P(B) = 0$ or $P(B) = 1$ then $P(B) > P(A)$. If $P(B) = 0$ then $m, n < 2$ so $P(A) = 0$ and thus $P(B) = P(A)$. For the case of $P(B) = 1$ then m or n is at least 2 and the other must be 0, thus $P(A)$ is also 1 and $P(B) = P(A)$.

For any other case where m or n is greater than 1 and the other is non zero, $P(B) > P(A)$

Intuitively, when you do not replace the ball, the probability you get the same color again is lower because there is less of the same color now. Therefore $P(A)$ is smaller than $P(B)$.

For a more formal proof, looking at the compliments of A and B, $P(A) = 1 - \frac{n*m}{(n+m)(n+m-1)}$ and $P(B) = 1 - \frac{n*m}{(n+m)^2}$.

It is clear that $(n+m)^2 > (n+m)(n+m-1)$ [when $n+m > 1$] so that means that $\frac{nm}{(n+m)^2} < \frac{nm}{(n+m)(n+m-1)}$

$\to -\frac{nm}{(n+m)^2} > -\frac{nm}{(n+m)(n+m-1)}$

$\to 1 - \frac{nm}{(n+m)^2} > 1 - \frac{nm}{(n+m)(n+m-1)}$

$\to P(B) > P(A)$

1.33

$|\Omega| = 6^5$

$|FH| = 6 * 5\binom{5}{3} = 300$

$P(FH) = \frac{300}{6^5} = \frac{25}{648}$

1.41

$A_i = \{\text{player } i \text{ wins no games}\}$

$P(A_i) = \frac{2}{3}^4 = \frac{16}{81}$

$A_i \cap A_j = \{\text{players } i, j \text{ win no games}\}$

$P(A_i \cap A_j) = \frac{1}{3^4} = \frac{1}{81}$

$A_1 \cap A_2 \cap A_3 = \{\text{no one wins any games}\}$

$P(A_1 \cap A_2 \cap A_3) = 0$ because at least one person must win each round.

$P(A_1 \cup A_2 \cup A_3) = \sum_{k=1}^{3}(-1)^{k+1} \sum_{1 \le i_1 < \ldots < i_k \le n} P(A_{i1} \cap \ldots \cap A_{ik}) = 3P(A_i) + 3P(A_i \cap A_j) + P(A_1 \cap A_2 \cap A_3) = \frac{16}{27} + \frac{1}{27} + 0 = \frac{17}{27}$

1.44 (tentative)

a

$X, Y \in \{1, 2, 3, 4, 5, 6\}$

b and c

k	1	2	3	4	5	6
$P(X = k)$	$\frac{1}{36}$	$\frac{1}{12}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$
$P(X \le k)$	$\frac{1}{36}$	$\frac{1}{9}$	$\frac{1}{4}$	$\frac{4}{9}$	$\frac{25}{36}$	1
$P(Y = k)$	$\frac{11}{36}$	$\frac{9}{36}$	$\frac{7}{36}$	$\frac{5}{36}$	$\frac{3}{36}$	$\frac{1}{36}$
$P(Y \le k)$	$\frac{11}{36}$	$\frac{5}{9}$	$\frac{3}{4}$	$\frac{8}{9}$	$\frac{35}{36}$	1